

210 FINAL 21

القسم الأول

- 1- The interrupt-request line is a part of the _____
 - a. Data line
 - b. Control line
 - c. Address line
 - d. None of the mentioned
- 2- Whenever the data is found in the cache memory it is called as _____
 - a. HIT
 - b. MISS
 - c. FOUND
 - d. ERROR
- 3- How can the processor ignore other interrupts when it is servicing one _____
 - a. By turning off the interrupt request line
 - b. By disabling the devices from sending the interrupts
 - c. BY using edge-triggered request lines
 - d. All of the mentioned
- 4-LRU stands for _____
 - a. Low Rate Usage
 - b. Least Rate Usage
 - c. Least Recently Used
 - d. Low Required Usage

5- In _____ mapping, the data can be mapped anywhere in the Cache Memory.

- a. Associative
- b. Direct
- c. Set Associative
- d. Indirect

6- The transfer between CPU and Cache is _____

- a. Block transfer
- b. Word transfer
- c. Set transfer
- d. Associative transfer

7- The circuit used to store one bit called

- a. Register
- b. Encoder
- c. Decoder
- d. Flip Flop

8- _____ register keeps tracks of the instructions stored in program stored in memory.

- a. AR (Address Register)
- b. XR (Index Register)
- c. PC (Program Counter)
- d. AC (Accumulator)

9- A register capable of shifting its binary information either to the right or the left is called a

- a. parallel register.
- b. serial register.
- c. shift register.
- d. storage register.

10- The register which contains the data to be written into or read out of the addressed location is called

- a. Memory address register
- b. Memory data register
- c. Program counter
- d. Index register

11- Example of an external interrupt is

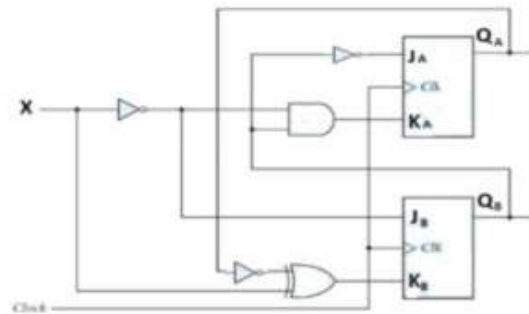
- a. Divide by zero interrupt
- b. Keyboard interrupt
- c. Overflow interrupt
- d. Memory error interrupt

12- The transfer between Main memory and Cache is _____

- a. Block transfer
- b. Word transfer
- c. Set transfer
- d. Associative transfer

القسم الثاني

Given SR or JK Flip-flop of the following block diagram



$J_A =$

$K_A =$

$J_B =$

$K_B =$

Current State		Input	Flip Flops Inputs				Next State	
Q_B	Q_A	X	J_A	K_A	J_B	K_B	Q_B	Q_A
0	0	0						
0	0	1						
0	1	0						
0	1	1						
1	0	0						
1	0	1						
1	1	0						
1	1	1						

القسم الثالث

$$X = (A * B - C) / (E * F + G)$$

Two-address	One-address	Zero-address

القسم الرابع

	Sequential tape	Direct Hard disk	Random Ram	Associative Cache
Give Example of each type				
Access time (1 to 2) 1: indicates fixed and 2 otherwise.	2	2	1	1
Speed (1 to 4) 1 indicate slowest and 4 indicates the fastest.	1	2	3	4
Locating data by address (1 to 2) 1: indicates locating data by address and 2 otherwise.	2	1	1	2
Locating data by parallel comparison (1 to 2) 1: indicates locating data by parallel and 2 otherwise.	2	2	2	1
Locating Data by search in order (1 to 2) 1: indicates locating data by search in order and 2 otherwise.	1	2	2	2
Locating Data by jump then little search in order (1 to 2) 1: indicates locating data by jump and little search in order and 2 otherwise.	2	1	2	2

جانا نفس الجدول بالحرف

القسم الخامس

Assume a processor using a cache with direct mapping; its data words are 8 bits long (i.e. 1 byte). The data addresses are to the word. If the physical address is 28 bits long and the tag is 11 bits. The block holds 16 bytes of data.

1) For such system calculate the following:

- a. The number of bits for word's field
- b. The number of blocks in the RAM
- c. The size of RAM
- d. Number of lines in the cache
- e. The size of cache

2) Consider the Address "A53C21F" and get the corresponding values of the different fields value in next table:

field	Tag	Line	Word
value			